Final report, Project Levante

Edge-AI Rapid Bacterial-Risk Detection for Mediterranean Bathing Water

ISEN Toulon Summer School 2026 — Challenge 4 — Team 3

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1 Executive Summary

Along the Mediterranean coast, millions of bathers are exposed each summer to bacterial contamination they cannot see. The problem is not that contamination goes undetected, but that it is detected too late: the standard laboratory culture takes between twenty-four and forty-eight hours to return a result, by which time swimmers have already entered the water. The consequences are both a public-health risk and a heavy economic cost to the coastal tourism economy that depends on open beaches. Levante addresses this delay rather than the measurement, treating the challenge as a problem of latency.

Levante is an edge-AI system that provides an early and continuous estimate of bacterial-risk for bathing water. A solar-powered buoy senses environmental proxies, chiefly turbidity together with rainfall, temperature and conductivity, and an on-board model converts these readings into a risk score in real time. When the risk rises, the system raises an alert that directs an accredited field team to the specific site, where a portable qPCR device confirms the result in roughly three hours. The guiding principle throughout is that prediction directs while measurement decides: the model never closes a beach, it only points the sampling effort to where it is most needed, and the closure decision remains with the accredited laboratory and the public authority. This design resolves the central tension between a bather who wants maximum caution and a business owner who wants no false alarm, because a mistaken prediction costs only a sampling trip.

The customer is the accredited testing laboratory, chosen because only an accredited result is legally admissible and because laboratories certify water quality but are not equipped to operate marine sensing at scale. The business model is circular: the buoys are leased rather than sold, so every unit returns at end of life for reuse, repair or recycling.

Within a two-week programme the team delivered a working turbidity sensing-and-classification node, built around an STM32 microcontroller and an on-device anomaly-detection model, and validated the optical separation between clear and turbid water. Against the current process, the proposed pipeline is projected to roughly halve the time between a contamination event and an actionable decision. The full buoy and the field-trained predictive model remain to be built, and the report sets out clearly what was achieved and what comes next.

2 Problem Statement

Challenge 4 of the programme, The Invisible Contamination, was stated in the brief as follows: every summer, beaches across the Mediterranean are closed because of bacterial contamination, yet the laboratory test that confirms the contamination takes up to forty-eight hours, by which time thousands of swimmers are already in the water. This section defines the problem the team chose to solve. Following the principle that a problem must be understood before any technology is proposed, it is centred around the people affected and the structure of the failure.

2.1 The interpretive decision that determined the project

The most consequential decision of the project was taken in its earliest hours, and every subsequent choice follows from it. On a first reading, Challenge 4 appears to be a detection problem, whose natural answer would be a more sensitive instrument. On closer examination, it proves to be a latency problem, which calls for a quite different response.

The distinction is not semantic, because the two readings lead to incompatible solutions. If the failure is one of detection, the goal is accuracy and the product is a superior sensor. If the failure is one of latency, the goal is time-to-decision, and accuracy that arrives late is worthless. The challenge itself supports the second reading: the standard culture method does not fail to detect contamination, it detects it correctly but slowly, because the indicator organisms require roughly twenty-four to forty-eight hours of incubation to grow to a countable quantity. The information is therefore not absent but delayed, and everything that follows in this report proceeds from treating the problem as one of delay rather than of measurement.

2.2 Context, challenge and opportunity

Context. The problem was addressed within a defined working setting: six teams of about six students each, deliberately assembled to be heterogeneous in background, working in parallel on a shared set of maritime and environmental challenges, with professors and mentors available for support. This setting is relevant to the problem statement because it shaped how the problem was analysed — collectively, from several disciplinary viewpoints at once, rather than by a single specialist.

Challenge. The challenge set as a whole concerned the Mediterranean sea and the human activity around it, a theme coherent with the maritime focus of the host institution and the surrounding industrial landscape. Challenge 4 sits within that theme as a public-health and coastal-economy problem rather than a purely technical one.

Opportunity. For the team, the opportunity was to learn how to approach a genuinely complex problem from two directions at once, the technical side and the economic-entrepreneurial side, while working inside a group whose members held different and complementary skills. The problem was therefore never treated as a device to be built; it was treated as a situation to be understood well enough that the right thing to build became clear.

2.3 The six dimensions of the problem

To characterise the problem precisely, it was decomposed along the six classical dimensions of who, what, where, when, why and how it is currently addressed. Table 1 summarises the result; the figures it cites are drawn from European regulatory and environmental sources.

Dimension	Characterisation
Who	Bathers and tourists, with children, the elderly and the immunocompromised at elevated risk; the public authorities responsible for sampling, classification and closure; beach concessions, hotels and restaurants whose revenue depends on the beach being open; and the wastewater utilities frequently identified as the root cause.
What	Microbiological contamination of bathing water, monitored through faecal indicator bacteria, where the operative failure is a structural detection delay of twenty-four to forty-eight hours rather than a failure to detect.
Where	European coastal bathing waters, with chronic problems concentrated in the Mediterranean states. Of the fifty-seven EU bathing waters classified "poor" for five consecutive years between 2020 and 2024, the large majority were in Italy and France [3]. The aggregate picture is reassuring, since roughly 85% of EU bathing waters are rated excellent, and this is precisely why the localised problem is underprioritised: the average conceals the peak.
When	Concentrated in the summer bathing season, with acute events triggered by intense rainfall producing combined sewer overflows, of which Europe has more than 650,000 [1]. Standard public-health advice to avoid bathing for one to two days after heavy rain is itself an admission that the monitoring system cannot resolve the event in real time. The trend is worsening: extreme daily precipitation intensifies by roughly 7% per degree of warming [11].
Why	Health: bathers face roughly double the odds of illness compared with non-bathers [5]. Economics: EU coastal tourism generated on the order of €93 billion in gross value added and around 2.7 million jobs in 2023 [2]. Regulation: the Bathing Water Directive requires member states to act on poor classifications, and five consecutive poor seasons trigger a permanent bathing ban [4].
How (today)	Four approaches, each closing a different gap and each incomplete: culture-based regulatory monitoring (accurate but slow and infrequent); rapid qPCR (fast but laboratory-bound and not the compliance method); portable biosensors (promising but still research prototypes); and machine-learning nowcasting (cheap and continuous but, in the published literature, missing roughly half of true exceedance events) [7, 8].

Table 1: The problem characterised along its six dimensions. Each existing approach in the final row optimises one axis (accuracy, speed, cost or coverage) at the expense of the others; none delivers all four at once.

2.4 Contamination has several sources, not one

An early version of the analysis treated rainfall and sewer overflows as the cause of contamination. A direct challenge to that assumption, namely are heavy rain and sewer overflows really the only causes?, corrected it, and the

correction changed the technical justification of the whole project. The literature identifies at least six distinct source categories, several of which operate in dry weather, as summarised in Table 2.

Source	Mechanism	Weather-driven?
Sewer overflows and discharges	Combined sewer systems exceeding capacity; continuous treatment-plant discharge	Partly
Agricultural runoff	Manure and animal waste transported by surface flow	Partly
Wildlife and birds	Direct faecal deposition near the shoreline	No
Sediment resuspension	Bacteria persisting in sand and sediment, remobilised by waves, boats or bathers	No
Bathers themselves	Shedding of indicator organisms during bathing, concentrated in the first minutes of immersion	No
Groundwater inputs	Beach-aquifer discharge driven by tidal cycles	No

Table 2: Contamination source categories. Because several pathways are independent of weather, a rainfall-based alerting rule alone is insufficient and continuous multi-variable environmental monitoring is genuinely required. The correction therefore strengthened the case for the sensing approach rather than weakening it.

This matters for the solution and not only for the statement. Were contamination purely rain-driven, a rule based on rainfall would suffice and no embedded sensing would be needed. Because several pathways are independent of weather, continuous multi-variable monitoring is necessary, a requirement that follows from the problem rather than from a prior wish to build a sensor. The sensing set adopted later, combining turbidity with rainfall, temperature and conductivity, targets the high-volume hydrological pathways in the upper rows of the table, the overflows, runoff and sediment resuspension that drive the most significant events. The diffuse biological sources in the lower rows, such as wildlife and bathers, remain the province of the scheduled accredited sampling that the system supplements rather than replaces. The device is therefore a screening layer for the events it can see.

2.5 Statement of the problem

Combining these dimensions gives the problem statement that governs the rest of this report:

> *Every year, in the Mediterranean sea, millions of people are affected by bacterial water contamination. This happens because the equivalent of 400,000 Olympic-sized swimming pools of raw sewage is spread into the waters, leaving no track or knowledge of where it goes.*
>
> *The situation is currently addressed through rare tests which do not keep up with the frequency of sewage disruptions. Moreover the usual time to get results is close to 48 hours. This creates economic losses for lack of proper planning which are estimated by an US firm to be between $1,300 to $37,000 daily for closed beaches, and there is not enough time to handle the situation in an ordered manner.*

This statement is the one carried into the pitch, and it deliberately leads with the scale of the problem and its economic consequence rather than with the technology. The daily-loss figures are attributed to their source, while the sewage-volume figure is reported as an illustrative order of magnitude rather than a precisely sourced quantity.

3 Making Sense of the Problem

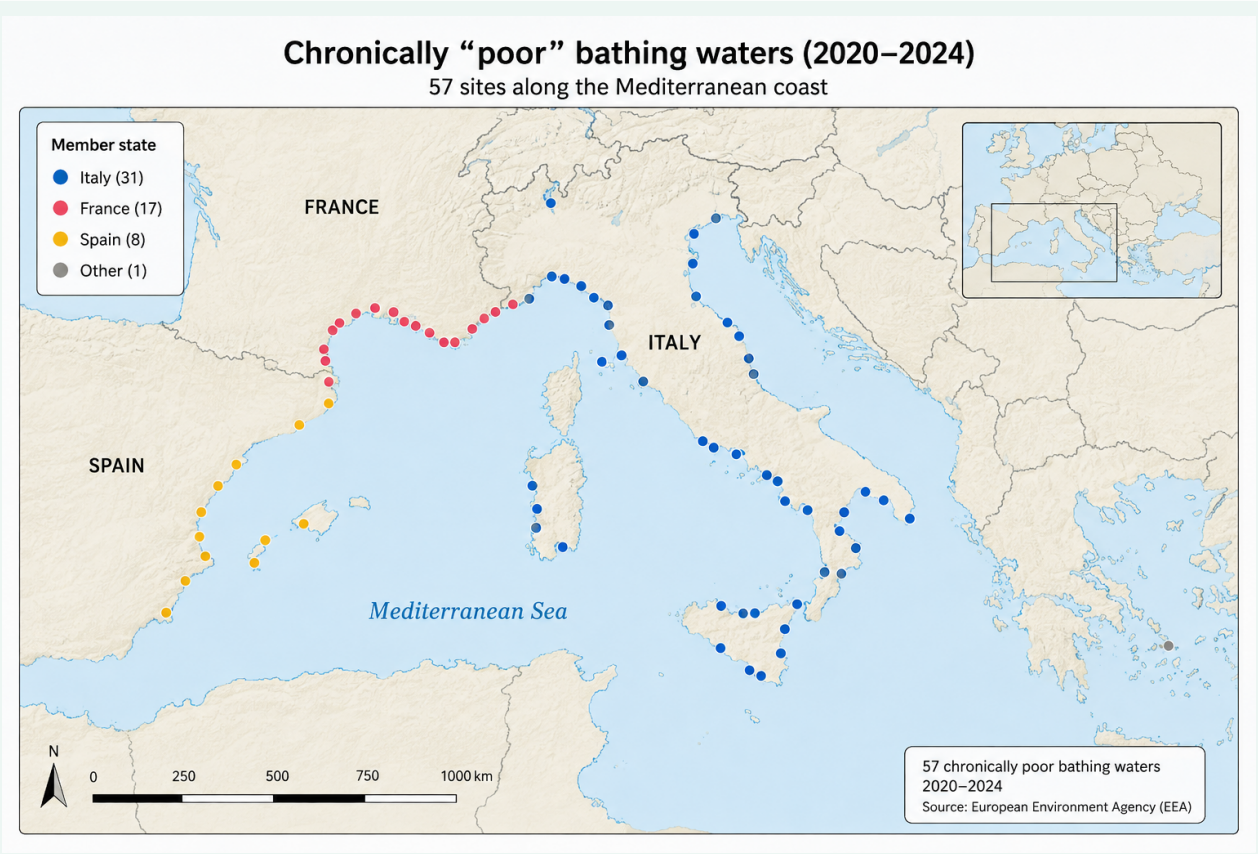

Figure 1: Geographic concentration of the problem. The chronically poor bathing waters cluster on the Mediterranean coasts, which is why an aggregate “85% excellent” figure conceals a real localised failure.

3.1 What sense-making is

Sense-making is the activity of taking a phenomenon, decomposing it into its constituent processes, understanding the people inside each process together with their needs, mapping the dependencies between those processes, and locating the pitfalls, both within a single process and, more importantly, at the joins between processes, along with the stakeholders who bear each one. The personas, needs and user journeys presented below were developed together, each feeding the others, and were finalised only once the whole cohered, then reworked a second time to remain coherent with the solution eventually chosen. This iterative co-production is itself the substance of the analysis.

The approach corresponds to what the organisational-theory literature calls sensemaking in the sense of Weick [9]: understanding is constructed retrospectively and socially, by acting on the situation and revising the account until it holds together, rather than by reading a fixed problem off the brief [10]. To that we add a complementary idea from process and value-stream analysis, in which waste and failure concentrate not inside individual activities but at the handoffs between them [6]. Naming these two lineages is not decoration: the first explains how the team built its understanding, and the second explains where the team looked for the failure once the processes were mapped.

The current process for detecting bathing-water contamination runs from periodic scheduled sampling, through laboratory culture, to the responsible authority's interpretation and finally the public signal that reaches a bather. Under the Bathing Water Directive, sampling is carried out only a handful of times per season at each site, on a fixed calendar rather than in response to events, and each sample is then grown in culture before it can be counted. The pitfalls the project targets all live at the joins of that chain, not inside its boxes: the long gap between one scheduled sample and the next, during which contamination goes unobserved; the twenty-four to seventy-two hour gap between taking a sample and obtaining a result, since bacteria must be grown before they can be counted; and the gap between a confirmed result and a warning reaching a bather before exposure rather than after. Figure 2 shows the chain with these join-level pitfalls marked; it is the single figure that makes the rest of this section legible as a whole rather than as two separate subsections.

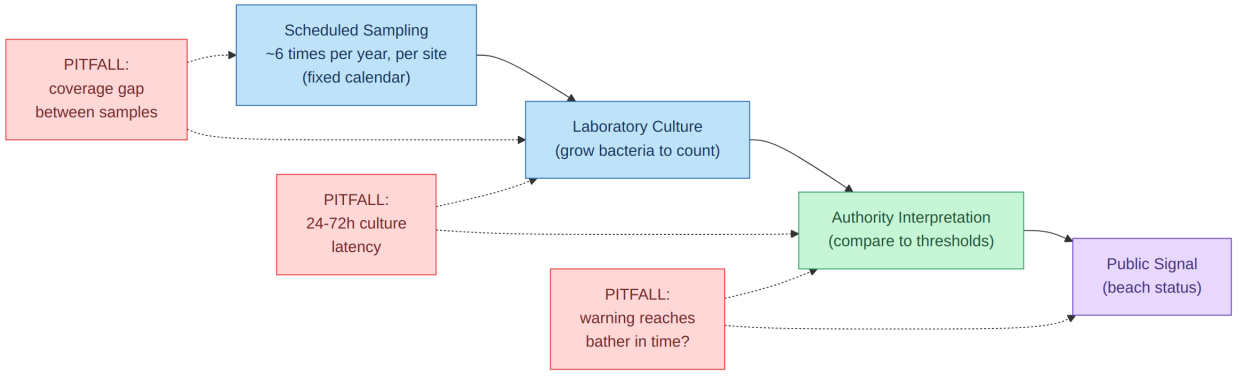

Figure 2: The problem as a chain of processes. The failures the project targets occur at the joins between processes rather than within them, which is why the solution is built around the joins.

3.2 Personas and user needs

Personas were used because they surface the key needs of a process by impersonating the people inside it. They were developed iteratively during the problem-understanding work and are deliberately fragmented, because they co-evolved with the user journeys rather than being written once in final form. In the method described above, that fragmentation is a natural property of genuine co-production.

Four personas were developed, each chosen for a distinct structural position in the problem rather than for demographic variety. Each occupies a different relationship to the contamination event, and a persona that merely duplicated another's relationship would have added colour without analytical value. Table 3 summarises them.

The four personas span the Mediterranean coast deliberately rather than a single town: three are situated on the French Riviera and one on the Catalan coast of Spain, reflecting the cross-border nature of the problem stated in Section 2.5 rather than any inconsistency of setting.

Persona	Structural position	Dominant need and frustration
Ping	Exposed, cannot act	Needs trustworthy confirmation that the water is safe before her children enter it; her frustration is helplessness, since contaminated water looks identical to clean water and she cannot judge it herself.
Gabriella	Accountable operator	Needs clear, well-organised data and a dashboard to monitor trends and justify decisions; her frustration is the weight of personal accountability for a call that may be wrong and is not fully within her control.
Josh	Field executor, no authority	Needs a clear route, a known priority site after an alert, and enough time to collect and return a sample before it degrades; his frustration is time pressure and distance.
Juan	Economically exposed	Needs predictable custom and protection of his reputation; his frustration is that poor water quality drives customers away and his business is blamed for it.

Table 3: The four personas, each defined by a distinct structural position in the problem: exposed to the event, accountable for detecting it, executing the detection, or economically damaged by it.

The needs conflict that defines the product. Two of the personas want opposing outcomes, and that tension is the hinge on which the whole solution turns. Ping wants maximum caution: she would rather be warned even when the evidence is uncertain. Juan wants the opposite: he does not want his season closed on a false alarm. The solution resolves this conflict by design rather than by compromise. A prediction from the machine-learning model never closes a beach; it only triggers an investigation, directing an accredited sample to the site it has flagged. The closure decision continues to rest on a certified physical sample and remains with the public authority. Because the prediction sets an investigation in motion rather than a consequence, a false prediction costs a sampling trip rather than a wrongful shutdown. On the question this raises directly, whether the business owner is told when the model flags his stretch of coast as risky, the answer follows from the same principle: Juan is notified only after the accredited result is available and the authority has acted, never on the strength of the prediction alone. He therefore learns of a confirmed closure, not of an unverified suspicion that could damage his trade without cause.

User needs. Collected from the personas, the needs the solution must satisfy are (i) a timely and trustworthy binary signal of whether the water is safe, delivered before exposure; (ii) the direction of scarce sampling effort to the site that most needs it; (iii) alerts that can be investigated for their cause rather than merely received; and (iv) certified, communicable evidence of safety that a business or an authority can act on and stand behind.

4 Proposed Solution

4.1 Choosing whom to solve the problem for

The most difficult decision in shaping the solution was not technical. The problem of Section 2 can, in principle, be solved for several different groups: the bathers who are exposed, the municipalities responsible for public safety, the laboratories that perform the analysis, or the coastal businesses whose revenue depends on open beaches. Each of these would lead to a different product. The decision of which to serve was taken deliberately, during the construction of the Business Model Canvas, and it determined everything about the solution that follows.

The decisive realisation came directly from the needs conflict identified in Section 3.2: the person who suffers the problem is not the person who can pay to solve it. Ping suffers the consequences of contamination but has neither the mandate nor the budget to commission a monitoring system. The party that both needs the capability and can pay for it is the organisation responsible for testing the water, that is, an accredited laboratory such as Eurofins, or the public authority that contracts one. The solution was therefore shaped around that customer, and the interests of the exposed public are served indirectly, through the customer who is paid to protect them.

4.2 Why the accredited laboratory is the right customer

Two arguments fix the choice. The first is legal and is a hard constraint rather than a preference: under the Bathing Water Directive [4], only a result produced by the accredited reference method is admissible for classification and closure, and the decision to close a beach rests with the public authority, not with any device or private party. Any solution that ignored this chain of authority would be inadmissible however good its measurements. The second argument is one of specialisation: laboratories are equipped and mandated to certify water quality, but they are not equipped to operate marine sensing hardware at scale. There is therefore a genuine division of labour to occupy: the team's system supplies continuous sensing and directed alerting; the laboratory supplies accredited confirmation; the authority supplies the decision.

This division also dissolves the Ping–Juan conflict cleanly. Because the laboratory and the authority retain all decision authority, the team's system never closes a beach. It only tells the laboratory where and when to look. A false alarm therefore costs a sampling trip, not a wrongful closure, which is precisely the resolution promised in Section 3.2.

4.3 The value the solution creates

Framed by that customer, the solution creates value along three lines, each traceable to a documented need rather than asserted. It directs sampling effort, so that scarce and costly technician trips are spent where contamination is most likely rather than on a fixed calendar, which addresses Josh's need for a known priority site and the laboratory's need to control operating cost. It shortens the certified turnaround, by pairing continuous prediction with portable confirmation, so that the customer can offer a faster service than the laboratory-bound status quo. And it provides continuous coverage between scheduled tests, closing the temporal blind spot that is the core of the problem. In the language of the Business Model Canvas, the value proposition is to increase testing efficiency and reduce operational cost while raising the granularity of monitoring through software. The customer sells a better, faster service, and the exposed public is protected as a direct consequence.

4.4 The end-to-end pipeline, from dataset to demonstration

The solution described so far is a system concept. To make it concrete, and to guide the two-week build, the team laid out the full pipeline from raw data to a working demonstration in three phases, shown in Figure 3. The figure is the bridge between the proposed solution and its technical realisation: its first phase is software, its second is hardware, and its third is the live demonstration by which the two were shown to work together.

Phase 1, machine-learning preprocessing and training, takes the raw open dataset (turbidity, salinity, weather and bacterial counts) and prepares it for a model. The bacterial targets are scaled to a single contamination index on a 0–100 scale; the physical inputs are mapped into the ranges the device's sensors produce; and the wind is decomposed into vector components. A regressor is then trained and converted to C code for deployment on the microcontroller.

Phase 2, hardware build and calibration, assembles the physical device and establishes the correspondence between real liquids and sensor readings, using three reference conditions (clear water, salted water, and a salted-milk demo mixture) to fix the boundaries the firmware uses.

Phase 3, the live demonstration sequence, is the scripted three-step demo: starting from clear water reading as safe, adding the demo mixture to drive the reading up, and finally switching the environmental context to show the index rising to its highest level. It is the demonstration that makes the abstract pipeline visible to an audience in a few minutes.

A note on scope. Figure 3 describes the intended, fully-featured demonstration pipeline, including a conductivity channel and a weather-context input alongside turbidity. As the results record (Section 7), what was physically completed within the two weeks was the turbidity sensing-and-classification node; the conductivity and context channels remained part of the demonstration design rather than the delivered build. The figure is therefore best read as the target the build was aimed at.

4.5 User journeys through the proposed solution

Having described the solution, its value can be seen most clearly by following the people who live through it during a contamination event. A user journey maps a single day for one persona, with timestamps, a mood score from one to ten, and the tools used at each step. These journeys are set within the proposed system: the alerts, the dashboard, the directed sampling and the confirmation step are all present, so the journeys double as a walkthrough of the solution in operation. Three viewpoints are traced: Ping the exposed bather, Juan the coastal business owner, and the operational pair of Gabriella and Josh who run the system.

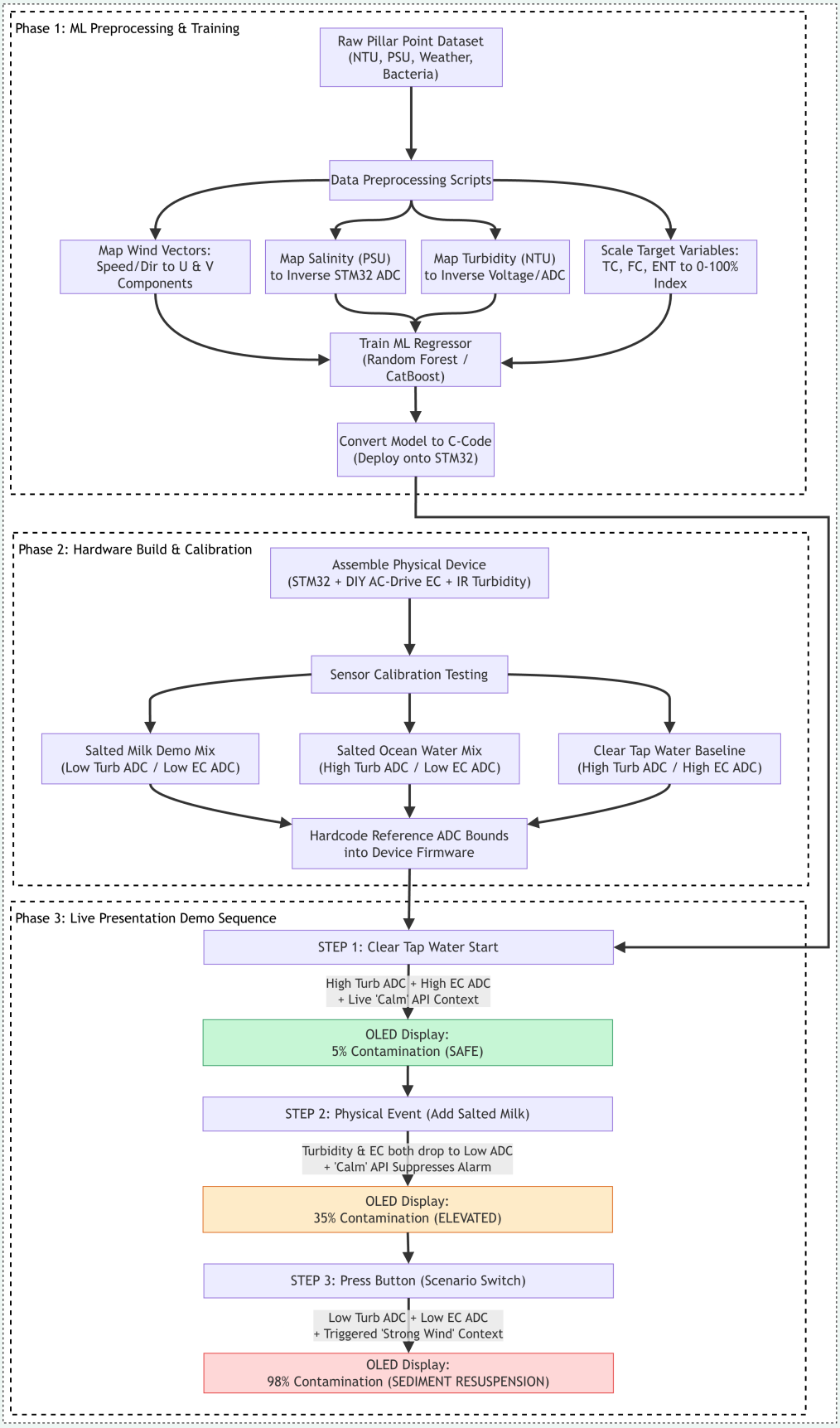

Figure 3: The end-to-end pipeline in three phases: machine-learning preprocessing and training (software), hardware build and calibration, and the live demonstration sequence. The diagram spans the full solution, from the raw open dataset to a scripted on-stage demonstration, and connects the proposed solution of this section to its technical realisation in Section 5.

Ping's journey. Table 4 follows Ping. Its defining feature runs against intuition: her mood rises at the moment she receives a notice that swimming is not allowed. For a user whose baseline experience is helplessness, a clear and authoritative answer is worth more than uncertainty, which is why the solution is designed to deliver a trustworthy signal that reaches her before exposure.

Time	What she does	Mood	Tools
Morning	Plans a beach afternoon with her two children; her old fear stirs, as they fell ill from bad water three years ago	5	Phone
15:55	At the beach, children ready to swim; her worry peaks at the water's edge	4	—
16:00	Receives a municipal SMS: swimming is not allowed today	7	Phone (municipal SMS)
16:05	Deep relief: this time she was warned before her children were exposed	9	Phone
16:20	Turns to a safe plan B, beach play with no water; her children are protected	8	—
Good day	No SMS; an on-site green flag shows safe water, the family swims and her fear finally eases	10	Phone / green flag

Table 4: Ping's journey through the system. Her mood rises on receiving a closure notice, because for a user whose dominant state is helplessness a timely authoritative answer is worth more than the absence of bad news.

Juan's journey. Table 5 follows Juan, the business owner. He receives an alert only once a closure is confirmed, and he uses the forewarning to adjust his orders and staffing rather than being caught unprepared. On a good day the absence of any alert, together with the on-site green flag, is itself the signal that his business can trade normally.

Time	What he does	Mood	Tools
Bad day, 08:00	Receives an alert that the water may be contaminated	3	Phone app
08:05	Adjusts the order to his suppliers and takes on fewer staff for the day	5	Phone
Good day	No SMS; the on-site green flag shows safe water and business runs normally	10	Phone / green flag

Table 5: Juan's journey. Forewarning of a confirmed closure lets him limit his losses; on a clear day the green flag signals business as usual.

The operational journey. Table 6 follows the people who operate the system, Gabriella and Josh, through the same event. The emotional shape is the mirror image of Ping's: Gabriella's mood falls across the day, reaching its lowest point not at the alert itself but at the moments of formal responsibility, when she notifies the authorities and signs the official report.

Time	Person	What they do	Mood	Tools
08:00	Gabriella	Starts her day working remotely; checks emails and the overnight data summary	8	Laptop, dashboard
08:10	Gabriella	Joins a short team meeting; reviews the sampling plan and staff schedule	7	Video call
08:15	Gabriella	Receives an alert of possible contamination, interrupting her management work	3	Notification system
08:20	Gabriella	Studies the dashboard to understand what caused the alert	5	Dashboard, data exploration
09:00	Gabriella	Notifies the local police and the town hall	1	Email
09:15	Gabriella	Briefs Josh and gives him the priority site to sample	4	Phone / email
09:30	Josh	Prepares the equipment and vehicle for the journey	3	Car, sampling kit
10:30	Josh	Arrives at the site	3	Car
10:40	Josh	Collects the sample and records the field conditions	3	Sample vial, field sheet
12:40	Josh	Returns to the laboratory and hands the sample to Gabriella	3	Car
12:45	Gabriella	Starts the analysis, then takes a break while it runs	3	qPCR machine
13:40	Gabriella	Reviews the model result and the context of the sample, then explores the data further	7	Dashboard, Python
15:40	Gabriella	Reaches a decision and prepares the official report	1	Report generator
16:00	Gabriella	Notifies the town hall so the beach can be officially closed	1	Phone

Table 6: The operational journey. Accountability concentrates on the operator: Gabriella's lowest points are the moments of formal responsibility, not the alert itself. On a good day both she and Josh see that all is well and continue their routine work.

What the journeys reveal. Placed side by side, the journeys show where the value and the cost of the system fall. Ping and Juan gain protection and forewarning, while the operators absorb the pressure of the decision so that the public does not. The mood scores make the point directly: they rise for the exposed users at the moment of a trustworthy warning, and fall for the operator at the moments of formal responsibility.

5 Technical Solution

5.1 Architecture: predict, then verify, then decide

The technical solution is organised as a chain that runs from continuous sensing to a human decision, shown in Figure 4. The design principle behind the whole chain is a strict separation of roles between its two halves: prediction directs, measurement decides. The left half of the chain, the buoy and its on-board model, is fast, cheap and continuous, but only ever produces a risk estimate; it is never permitted to trigger a consequence on its own. The right half, portable confirmation and the authority's decision, — is slower and more expensive, but authoritative.

Each half is used only for what its failure mode allows: a wrong prediction wastes a trip, whereas a wrong closure would harm a business or a bather, so closure is never placed on the predictive half.

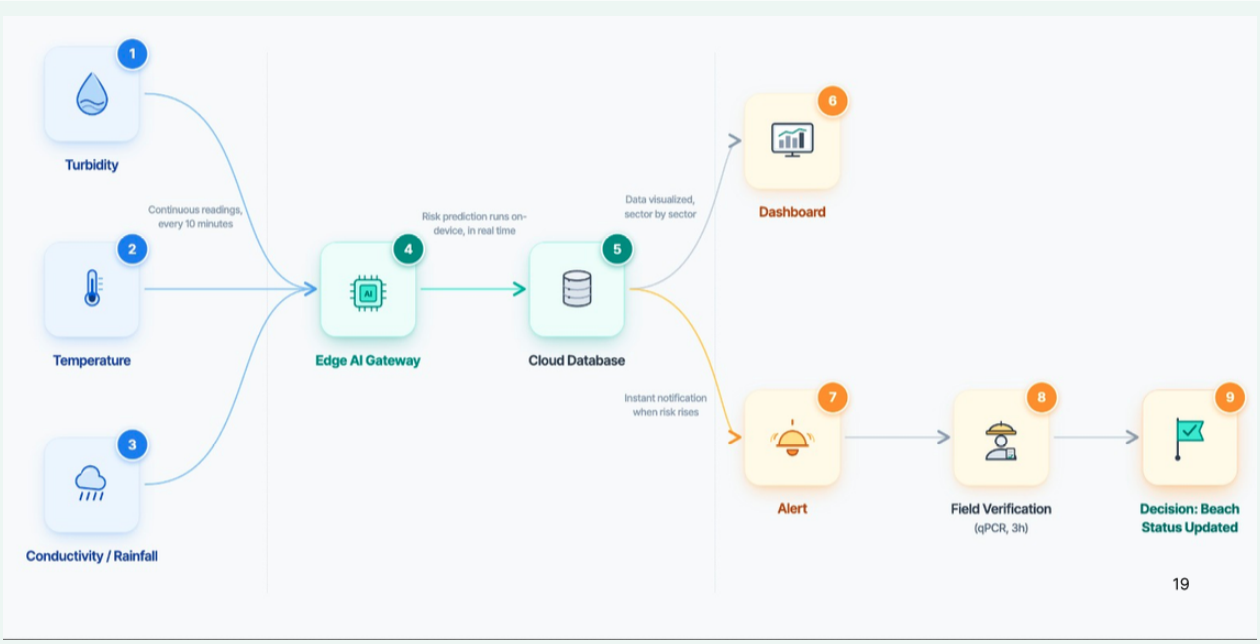

Figure 4: The Levante architecture as a nine-stage chain. Continuous environmental readings (1–3) are fused and classified on-device by the Edge-AI gateway (4); the resulting risk estimate is stored and visualised sector by sector (5–6); when risk rises, an alert (7) directs a field team to the specific site for portable qPCR confirmation (8); and only then does the responsible authority update the beach status (9). Prediction occupies stages 1–7 and directs; measurement and decision occupy stages 8–9 and decide.

5.2 The two solution families and their importance

Underlying the architecture is the recognition that there are two fundamentally different ways to attack the latency problem of Section 2.1, each with a complementary weakness. One can measure faster, replacing the slow culture with a rapid method such as qPCR, which closes the latency gap directly but remains an expensive, discrete test that says nothing between samples. Or one can predict, inferring contamination risk continuously from cheap environmental proxies, which closes the frequency gap for almost no marginal cost, but inherits the published limitation of nowcast models, which miss on the order of half of true exceedance events [8]. Neither family is adequate alone: rapid measurement is too costly to run continuously, and prediction is too unreliable to act on directly. The architecture uses each only where its weakness is tolerable: prediction to direct attention cheaply and continuously, measurement to decide accurately and occasionally. It is this pairing, rather than either technology in isolation, that constitutes the solution.

5.3 Sensing: the buoy

The sensing layer is a solar-powered buoy carrying a set of environmental sensors read at ten-minute intervals. The parameters were chosen from the requirements of the predictive model, not from the availability of sensors: the nowcasting literature consistently identifies turbidity, rainfall, water temperature and conductivity or salinity among the strongest proxies for faecal indicator bacteria [7, 8]. Table 7 lists the subsystems. Turbidity is the parameter the team implemented physically, as described in the results.

5.4 Edge AI: computing the risk on the buoy

Inference runs on the buoy, not in the cloud. The on-board microcontroller converts each set of readings into a risk estimate locally, and only that estimate, a small quantity, is transmitted. This decision is justified on two grounds that reinforce each other: bandwidth and power. Transmitting a continuous stream of raw multi-sensor data from an unattended solar-powered buoy would be costly in both, whereas transmitting a periodic risk score and its supporting context is cheap enough to sustain on harvested solar energy. Placing the intelligence at the edge is therefore not

Subsystem	Role
Optical turbidity	Primary proxy for suspended matter and, indirectly, for contamination; the parameter physically prototyped by the team.
Temperature probe	Water temperature, a driver of bacterial survival and a model input.
Conductivity	Salinity/conductivity, discriminating freshwater intrusion (e.g. runoff, outfalls) from open seawater.
Rain gauge	Precipitation, the trigger for overflow-driven events.
Current / dynamics	Local water movement, informing transport of a contamination plume.
Edge-AI gateway	On-board microcontroller running the inference that converts readings into a risk estimate.
Telemetry link	Low-bandwidth uplink transmitting the risk estimate and context.
Solar power	Energy autonomy for unattended deployment.

Table 7: Buoy subsystems. The sensing parameters follow from the predictive model's strongest inputs rather than from what happened to be available.

a fashionable choice but a consequence of the deployment constraints, and it is what the low-power embedded inference tool used by the team is designed for.

5.5 Confirmation, dashboard and communication

When the risk estimate crosses a threshold, the system raises an alert that directs a field team to the specific site, where a portable qPCR device produces a confirmation result in approximately three hours rather than the one to two days of laboratory culture. The portable qPCR is supplied through a hardware partner rather than developed in-house, which keeps the team's contribution focused on the sensing-and-prediction layer where its value lies. Because sampling is now directed by the prediction rather than performed on a fixed schedule, the total number of tests required falls even as coverage improves.

Two supporting components complete the chain. A cloud database and dashboard present the risk estimates sector by sector, specified from Gabriella's profile in Section 3.2: an operator console, not a consumer application, whose purpose is to let an accountable professional monitor many sectors at a glance and investigate the cause of an alert through the underlying time series. The communication layer follows a single rule that recurs throughout the design: the result travels outward and context travels inward, but raw signal never leaves the buoy, which is the same principle that justified on-board inference in Section 5.4.

6 Project Management

6.1 Team and division of labour

The team comprised six members of deliberately mixed background. The working method followed directly from that composition: for the first block of work — understanding the problem, the whole team worked together, so that no individual could produce analysis that drifted incoherent with the rest. Only once the problem was jointly understood did the team split by skill for the parallel work of building, research and documentation. Table 8 records the division. The pairing of the joint phase followed by a skill-based split is itself a managed decision, and it is the reason the many artefacts of Section 3 remained mutually consistent.

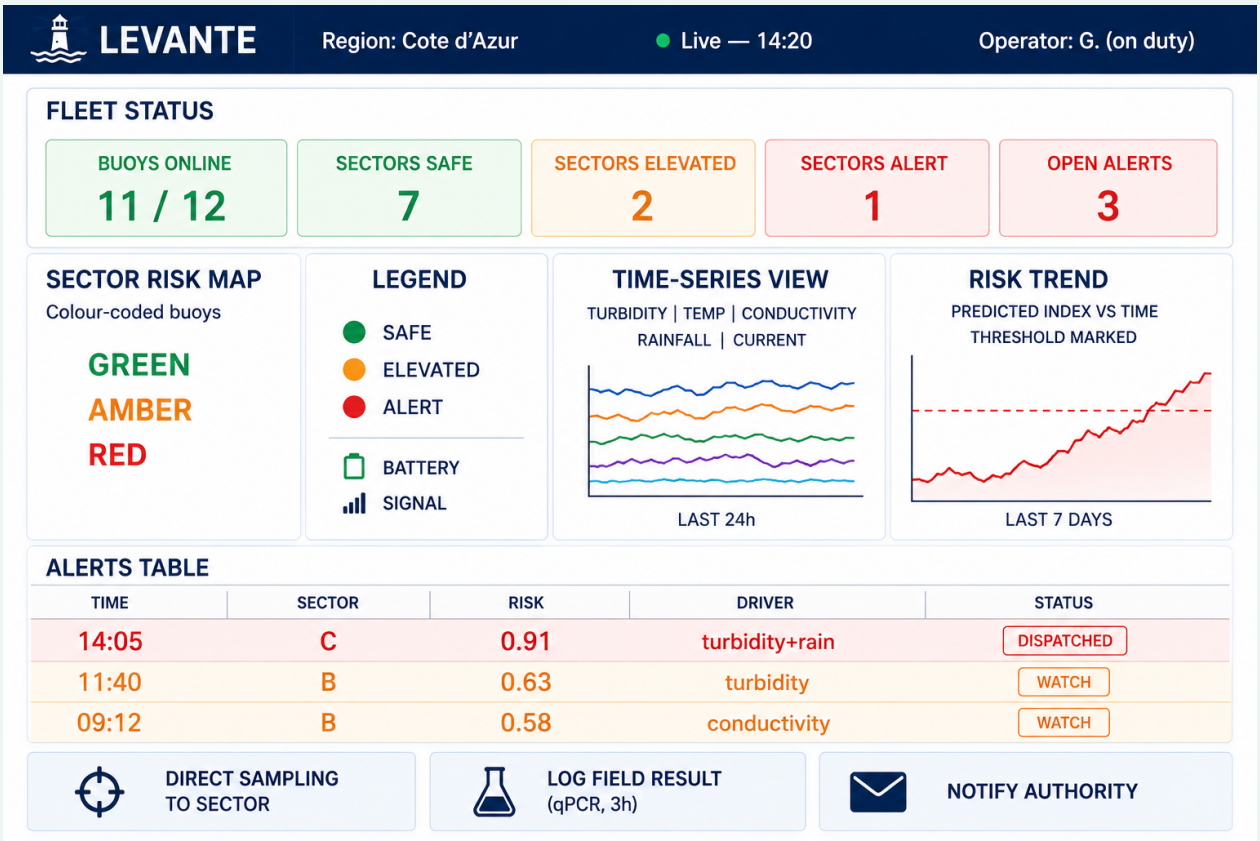

Figure 5: The operator dashboard concept. It is designed for Gabriella's accountability profile: many sectors monitored at once, and every alert traceable to its underlying cause through the time series.

Member	Principal contribution
Riccardo Cirillo	Entrepreneurship, economics and effective communication; business and circular canvases, pitch.
Victor G. De Matteis	Autonomous systems and entrepreneurship; research, documentation and overall direction of the project.
Raul Frandes	Computer science and artificial intelligence; machine learning and embedded-systems programming.
Carolina Mata	Technical work, in particular the design and implementation of the electronic circuits.
Giorgio Perakis	Mechanical design, critical sense-making of information and embedded-systems programming.
Peng Ying	Graphics, the display and arrangement of information, and the production of professional documents.

Table 8: Division of labour. The team worked jointly on problem understanding, then split along complementary skills for the parallel build, research and documentation work.

6.2 Timeline and intermediate deliverables

The project ran over two weeks. The first week was devoted to understanding the problem and to the analytical artefacts (personas, user needs and journeys), and closed with an intermediate presentation. The second week was devoted to the business and circular canvases, the embedded-AI and turbidity-sensing prototype, the storyboard, and the final jury pitch. The intermediate presentation at the end of the first week served as a genuine checkpoint: it forced the analytical work to cohere into a single narrative before any substantial build began, and it surfaced several inconsistencies early enough to correct cheaply. The full artefact inventory (problem statement, four personas, three journeys, storyboard, two canvases, architecture definition, the turbidity prototype and the final deck) was assembled across the two weeks and is listed in the results.

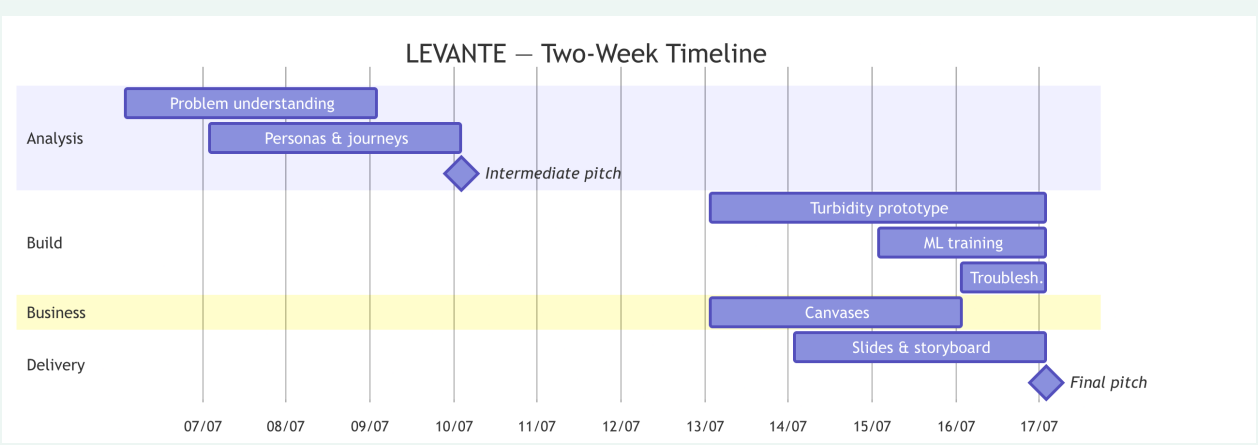

Figure 6: Two-week project timeline. Week one converges on the intermediate pitch; week two runs the build and business workstreams in parallel to the final pitch.

6.3 Method: how conclusions were reached

The soundness of the conclusions rests on the method by which they were reached, and that method is set out explicitly here. Four disciplines were applied throughout. Claims were held to an evidence standard: an empirical statement was either cited or explicitly flagged as unverified rather than presented as established fact. Assertions were made search-first, checking a source before stating a number rather than after. Design proceeded by iterative refinement, weighing each option against the constraints of the problem before committing to it. And, as the number of artefacts grew, cross-artefact consistency was checked deliberately across documents rather than within them, because the dominant risk had shifted from producing content to producing contradictory content. This last discipline surfaced most of the errors recorded in Section 6.4, including the two join-level inconsistencies already discussed in Section 4.5. Throughout, the team made use of AI assistants, namely Claude, Gemini and ChatGPT, to accelerate background research and to cross-check technical detail, while retaining responsibility for every figure and claim carried into the report.

The points at which the project changed direction, and the reason for each, are recorded in the decision log of Table 9. The log is included deliberately: a decision that cannot state the option it rejected and why has not genuinely been made.

6.4 Challenges encountered and resolved

The challenges fell into two groups. The analytical challenges were resolved by decision rather than drift: an early architecture in which two incompatible devices coexisted was collapsed to one (buoy deploys, model triggers, portable qPCR confirms); a legally incorrect claim about who holds closure authority was corrected, which strengthened rather than weakened the argument; and an over-length final deck was cut by consolidating the storyboard and demoting the forty-cell canvas to a brief credibility flash.

The hardware challenges, encountered while building the turbidity node, were more numerous and are the clearest record of the practical engineering carried out. Table 10 records the principal ones with their resolutions. They are characteristic of embedded development: each symptom was misleading, and each fix required diagnosing the layer at which the real fault lived rather than the layer at which it appeared.

#	Decision point	Chosen	Rationale
1	Interpretation of the challenge	Latency, not detection	Culture detects correctly but slowly; a biological constraint, not a technical failure.
2	Solution family (a broad class of approach to the problem)	Both, in defined roles	The two classes, faster measurement and prediction, close different gaps and fail differently, so each is used where its failure mode is tolerable.
3	Deployed device	Sensing buoy	Continuous coverage addresses the frequency gap; qPCR reserved for confirmation.
4	Role of prediction	Sampling director	Published nowcasting sensitivity ($\approx$50%) is unacceptable as a closure trigger.
5	Primary customer	Accredited laboratory	Accreditation is the binding legal constraint; the specialisation argument holds.
6	Commercial model	Service (lease + fee)	Enables take-back circularity and recurring revenue; matches sector norms.
7	Prototype scope	Turbidity-sensing node	Validate the least-obvious component first; the buoy platform is low conceptual risk.
8	Prototype implementation	Staged (safe level first)	Guarantees a working demonstration regardless of time overrun.
9	Machine-learning model	Developed by a partner	Training a field-ready model was beyond a two-week scope, and the choice is stated openly in the business model.
10	Personas shown in the pitch	Ping and Gabriella	These two sit at opposite ends of the same chain; Josh and Juan appear in the storyboard instead.
11	Canvas in the presentation	Shown briefly, then summarised	A forty-cell canvas cannot be read when projected, so only its essence was presented.

Table 9: Decision log. Each row records a point at which the project changed direction, the option chosen, and the reason the alternatives were rejected.

Symptom	Diagnosis and resolution
ADC returning near-zero on every pin	The analog supply was never enabled; the required call to enable it is not generated automatically and had to be added by hand before the ADC would convert.
Build failure, implicit declaration of an ADC function	A HAL driver module was not active; enabling it in the configuration restored the missing declarations.
GDB server fails to start on a fixed TCP port	An orphaned debug-server process was holding the port; it was terminated, and the sequencing of the tools sharing the debug probe was corrected.
Debug probe unreachable after a low-power sleep state	On-board sleep had halted the debug interface; a full flash mass-erase recovered the board.
Probe rejected after a hardware swap	A hard-coded probe serial number no longer matched; disabling the fixed serial and rescanning restored deployment.
Primary IR phototransistor failure	The optical sensor failed during final testing; a replacement analog sensor was sourced, with the consequence that the model would require retraining because a different sensor has a different spectral response.

Table 10: Principal hardware challenges and their resolutions. In each case the observable symptom pointed at a different layer from the actual fault, which is the normal texture of embedded development.

6.5 The human side of two weeks

An engineering report is also an account of a team, and the human experience of the two weeks is part of the record. In the first week the team built good working relationships by working together on the problem; in doing so, the members came to understand one another's points of view, and the intermediate deliverable strengthened those relationships further by mapping each person's competences and building trust.

The second week was more demanding, particularly during the prototype build. After developing the parts of the embedded system in parallel, merging them proved difficult, and the difficulty produced some genuine tension, most of it arising during the troubleshooting of the sensing board, where four members effectively stopped other work to concentrate on a single stubborn fault. Those tensions were worked through rather than left to fester, and the team recovered its coordination in time to prepare a strong final pitch. By the end, every member felt satisfied with the result and felt that they had actively and genuinely contributed to it — which, given the mixed backgrounds and the time pressure, is the outcome the team is most pleased with.

7 Results

7.1 What was built

Two weeks, of which roughly ten hours were available for hardware, do not permit building a marinised autonomous buoy. The team therefore made an explicit scoping decision, recorded as decisions 7 and 8 in Table 9: build the one component whose feasibility was least obvious, the optical turbidity sensing-and-classification node, and design the rest. What was physically delivered is a working turbidity node comprising a 3D-printed stand holding an infrared emitter and receiver as an optical cross, with an integrated cup-holder that positions the sample vessel correctly and repeatably in the beam, the drive circuit for the (continuously on) emitter, the transimpedance and analog-to-digital front end by which the microcontroller reads the receiver, and a NanoEdge AI classification model trained on data gathered with the rig itself. Given the time budget, a single emitter–receiver beam was implemented rather than the full cross.

7.2 The data-gathering rig and its discipline

The measurement principle is optical transmissivity: the emitter shines infrared light through the liquid, suspended particles scatter and absorb it, and the light reaching the receiver is inversely related to turbidity. The rig was

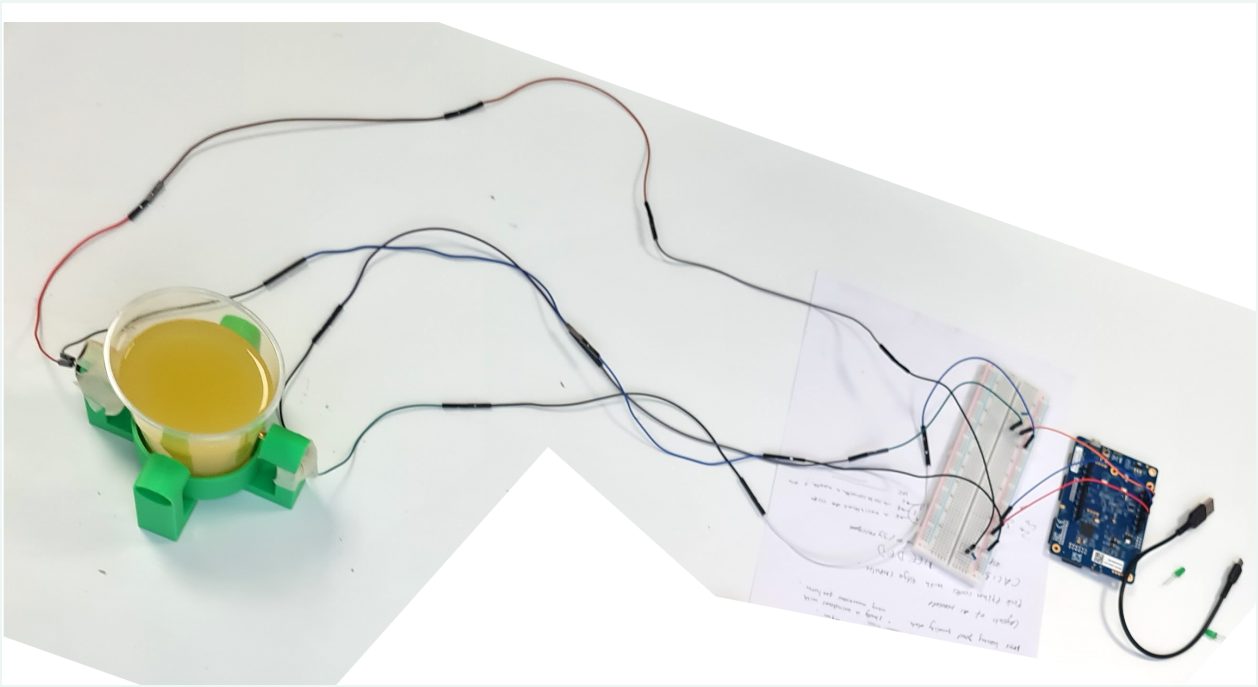

Figure 7: The turbidity sensing node. The emitter and receiver are held rigidly in the 3D-printed stand on either side of the sample cup, so the optical path cannot shift between readings; the wiring runs through a breadboard to the microcontroller, and during measurement an ambient-light box (not shown) covers the cup to exclude ambient infrared. The cup here holds orange juice rather than the milk dilutions used later, as this was one of the first tests carried out and is the only photograph available of the rig in operation.

built to protect that measurement from two contaminating influences. The emitter and receiver were held rigidly facing each other so that the optical path could not shift between readings, since a shift would silently invalidate the calibration, as noted below. The sample glass was placed in a holder between them, and a box was placed over the whole assembly so that the receiver measured the emitter's light net of ambient light rather than daylight. This dark-reference discipline matters because ambient infrared enters the measurement as a bias, and an uncorrected bias cannot be removed after the fact.

A second discipline was applied in firmware. Passing single raw ADC readings to the classifier proved too noisy, because the electrical noise floor was large relative to the difference between classes. Each reported value is therefore the average of many rapid conversions, which suppresses the noise while preserving the signal. This is the one place in the pipeline where taking more samples genuinely helps.

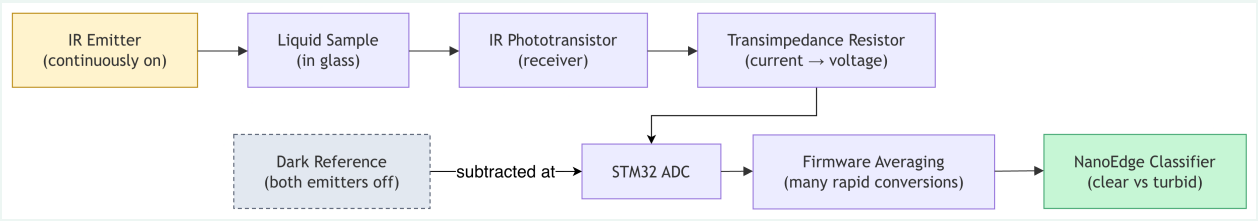

Figure 8: The sensing signal chain, from emitted light to class decision. Noise is suppressed by averaging at the ADC stage, and ambient light is removed by dark-reference subtraction.

7.3 The classification model

Two classes were collected by submerging the probe in controlled conditions. Clear water (tap water) established the baseline of maximum optical transmission and therefore the highest ADC readings. Turbid water, the anomaly class, was produced by adding milk in varying amounts: the suspended lipids and proteins scatter infrared strongly, sharply reducing the light reaching the receiver and so the ADC reading. Data were collected over several minutes per class to capture slow drift, and NanoEdge AI Studio then selected and trained a model small enough to run within the microcontroller's memory, learning the statistical distribution of each class and the boundary between them. The library was exported for integration into a STM32CubeIDE program.

The honesty of what this model is, and is not, must be stated. Milk is a convenient way to make water optically turbid, but it is not a traceable turbidity standard: its particle-size distribution differs from the reference material that defines nephelometric turbidity units, so the model distinguishes clearer from more turbid reliably but does not report calibrated turbidity values. The model is also, by construction, tied to the exact physical setup: any change in the emitter-to-receiver distance or in the transimpedance resistor invalidates the trained boundary and requires re-training. Both facts are limitations of the two-week prototype, not of the approach, and are stated because naming them is what distinguishes a result from a demonstration.

7.4 Prototype test observations

The two classes separated cleanly, and the model's behaviour was checked with the emulator before the library was exported. The emulator runs the candidate model on recorded signals on a workstation, so that the algorithm can be verified to work before it is committed to the device; it gave good results here. To quantify the separation independently of NanoEdge, the recorded signals were also analysed directly. Three conditions were logged: clear water (the good class) and two anomaly levels produced by adding twelve and twenty-four drops of milk — and the mean infrared reading of each was compared, then fitted with an ordinary least-squares regression of received signal against the number of drops.

The means fall monotonically and substantially with turbidity, and the linear fit is strong, confirming that the optical signal carries the turbidity information the classifier relies on:

The regression over all samples gives a coefficient of determination of $R^2 = 0.956$, with the fitted relationship

$$\text{ir_received} = 3920.17 - 45.64 \times \text{drops_water},$$

that is, each added drop of milk reduces the received signal by about 46 counts. The strength of the fit ($R^2 = 0.956$) confirms that turbidity, as emulated by milk concentration, maps to the optical reading in a stable and predictable way over the tested range, which is exactly the property the anomaly classifier depends on. The analysis script is reproduced in Listing 1.

Condition	Drops	Mean IR received
Clean water	0	3877.2
12 drops	12	3458.4
24 drops	24	2782.0

Table 11: Mean infrared reading for the three logged conditions. The signal falls by roughly 1100 counts from clean water to the strongest anomaly, a clear and monotonic separation.

Listing 1: Regression analysis of the exported signal files (clean water versus two milk-anomaly levels), used to verify the optical separation independently of the NanoEdge model.

```
import pandas as pd
import statsmodels.api as sm
import numpy as np

files = {
    "CleanWater.csv": 0,
    "12drops.csv": 12,
    "24drops.csv": 24
}

X_data = []
y_data = []

mean_table = []

for filename, drops in files.items():

    # Load CSV
    data = pd.read_csv(filename, sep=r"\s+", header=None)

    # Flatten all signals and all time samples
    values = data.values.flatten()

    # Store all samples for regression
    y_data.extend(values)
    X_data.extend([drops] * len(values))

    # Mean for display
    mean_table.append({
        "file": filename,
        "drops_water": drops,
        "mean_ir_received": np.mean(values)
    })


# Table of means
mean_df = pd.DataFrame(mean_table)

print("\nMean values:")
print(mean_df.to_string(index=False))


# Regression
X = pd.DataFrame({"drops_water": X_data})
y = pd.Series(y_data)

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()


print("\nRegression:")
print(model.summary())


# Equation
b0 = model.params["const"]
b1 = model.params["drops_water"]

print("\nRegression equation:")
print(f"ir_received = {b0:.4f} + ({b1:.4f}) * drops_water")
```

7.5 Outcome against the challenge

Measured against the challenge, the twenty-four-to-forty-eight-hour confirmation delay, the proposed process changes the timeline as summarised in Table 12. The headline figure is a reduction of approximately 50% in the time between a contamination event and a legally actionable decision. This figure is a projection derived from the documented portable-qPCR turnaround and the user-journey timeline of Section 4.5, not a measurement from a deployed system, and it is presented as such.

Stage	Current process	Levante
Detection of a possible event	Only at scheduled sampling	Continuous, every ten minutes
Prompt to investigate	None until a scheduled sample	Immediate on-model alert
Sample collection	Scheduled	Directed to the priority site within hours
Confirmation	24–48 h laboratory culture	≈3 h portable qPCR

Table 12: The proposed process against the current one. The confirmation step is shortened and, more importantly, the detection and prompt steps become continuous rather than scheduled.

7.6 The circular business model

The entrepreneurial track produced a second result alongside the prototype: a worked-out business model, expressed as a Circular Business Model Canvas (Figure 9). It is included among the results because it is what positions the technical work in a market; an engineering solution without a viable commercial model is only half a solution.

What the canvas contains. The canvas states the mission, to reduce the structural delay between real contamination and its detection in bathing waters, and builds a coherent operating model around it. The value proposition is to increase testing efficiency, reduce operational cost, and raise monitoring granularity through software. The users and contexts are private accredited laboratories and those who use the sea, and the revenue comes from a service fee charged to the laboratory (rather than a hardware sale), supplemented by licensing of the environmental dataset the system accumulates and an optional dashboard subscription. The partners are named concretely: a microcontroller manufacturer for the hardware, a portable-qPCR maker for the confirmation step, an external provider for the machine-learning model, and EU funding bodies for the pilot. The costs are the on-site testing equipment, the field sampling and analysis staff, and the maintenance of the technology stack.

Why it is a circular canvas. The model was built on a circular template because the hardware is physical and distributed, and its end of life matters. The decisive commercial choice is that the buoys are leased, not sold: ownership never transfers, so every unit returns to the company at end of life, which turns a low ambient recovery rate into near-total controlled return and aligns the environmental incentive with the commercial one. The canvas records the corresponding end-of-use provisions: modular components for separate recycling, a user-replaceable battery, long-term microcontroller availability for repair, and the reuse of retired units to calibrate nearby active sensors. It also states the trade-offs honestly, in its own negative-impact row: the manufacturing and energy footprint of distributed electronics, the critical materials in batteries and boards, and the risk of a false sense of security if the system were ever treated as a replacement for accredited laboratory analysis rather than a complement to it.

Market position and segment. Taken together, the canvas positions Levante not as a testing device sold to the public or to municipalities, but as a service to the accredited testing sector, the business-to-business segment identified in Section 4. This is a deliberate market choice: it places the product beside the laboratory's existing accredited workflow rather than in competition with it, and it sells continuity of monitoring and direction of effort rather than a one-off measurement.

Connection to the technical project. The canvas and the technical solution are two views of one system. The technical resources block names exactly the components of Section 5: the environmental sensors, the microcontroller board, the portable qPCR device and the cloud dashboard, and the key activities block names the work the engineering must sustain: sensor development, model training and updates, field calibration, and on-site confirmation. The leasing decision, in turn, imposes technical requirements back on the design: modularity, a replaceable battery and long-term repairability are engineering constraints that exist because of the circular business model. The two tracks were developed in parallel precisely so that each would constrain the other into coherence, and the canvas is the artefact where that coherence is made visible.

7.7 What remains to achieve

Setting out clearly what is still to be done is part of an honest result. The complete buoy exists as a design rather than a built device, and building it is the natural next stage. The predictive model has not yet been trained on Mediterranean field data. The sensitivity figures drawn from the literature come from other coasts, and because nowcast performance depends strongly on the site, the model will need local training and validation before it can be deployed. The 50% improvement is a projection rather than a measurement taken in the field, and confirming it in a real deployment is a clear objective. Costs were compared against published figures rather than quoted by a supplier, and no commercial agreement with an accredited laboratory was reached within the two weeks. None of these open points weakens the result. Each one marks the line between what has been demonstrated and what comes next, and that line has been drawn on purpose.

8 Conclusion

The project delivered a coherent chain from evidence to architecture to commercial model, built on the single interpretive decision that the problem is one of latency rather than detection. From that reading followed everything else: a solution that pairs continuous cheap prediction with occasional accurate confirmation, a customer chosen because accreditation is the binding constraint, and a working prototype of the one component whose feasibility was least obvious. The most useful output of the two weeks is not the device but the reframing — moving the moment of knowing to before the moment of harm rather than after it.

The way the team worked is inseparable from what it produced. Every member contributed actively to the result; the parallel workstreams were kept coherent by deliberate cross-checking rather than by luck; and the analysis proceeded iteratively, revisiting the problem and the solution from several viewpoints until they held together. The difficulties of the second week, when independently developed parts had to be merged under time pressure, were real and were worked through rather than avoided.

9 References

References

[1] EurEau. Combined sewer overflows in europe. Technical report, EurEau, 2020.

[2] European Commission, DG MARE. The EU blue economy report 2025 — coastal tourism. Technical report, Publications Office of the European Union, 2025.

[3] European Environment Agency. European bathing water quality in 2024. Technical report, European Environment Agency, 2025.

[4] European Union. Directive 2006/7/ec concerning the management of bathing water quality, 2006. Bathing Water Directive.

[5] A. F. C. Leonard, A. Singer, et al. Is it safe to go back into the water? a systematic review and meta-analysis of the risk of acquiring infections from recreational exposure to seawater. International Journal of Epidemiology, 47(2):572–586, 2018.

[6] M. Rother and J. Shook. Learning to See: Value Stream Mapping to Add Value and Eliminate Muda. Lean Enterprise Institute, Cambridge, MA, 1999.

[7] R. T. Searcy, A. B. Boehm, et al. Implementation of an automated beach water quality nowcast system at ten california oceanic beaches. Journal of Environmental Management, 223:636–645, 2018.

[8] J. Vidal et al. A critical review of model construction and performance for nowcast systems for faecal contamination in recreational beaches. Science of the Total Environment, 954, 2024.

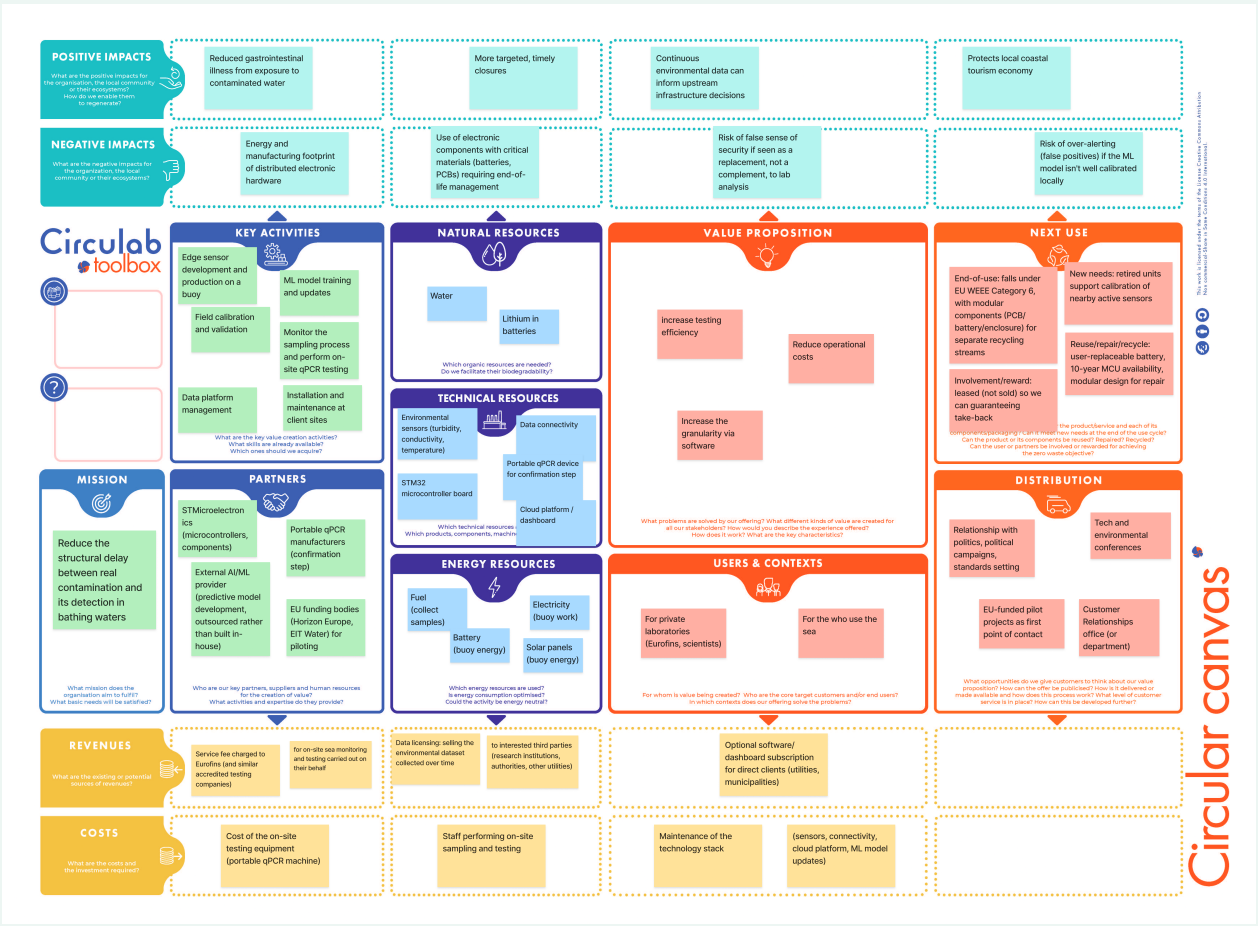

Figure 9: The Circular Business Model Canvas. Beyond the classic value, customer, revenue and cost blocks, the circular template adds explicit positive- and negative-impact rows and a “next use” block, which together record the leasing-and-take-back model, the end-of-life provisions, and the stated environmental trade-offs.

[9] K. E. Weick. Sensemaking in Organizations. SAGE Publications, Thousand Oaks, CA, 1995.

[10] K. E. Weick, K. M. Sutcliffe, and D. Obstfeld. Organizing and the process of sensemaking. Organization Science, 16(4):409–421, 2005.

[11] G. Zittis et al. Revisiting future extreme precipitation trends in the Mediterranean. Weather and Climate Extremes, 34:100380, 2021.